Open problems:
Non-linear representations in quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Density operators and entropy on $m(H)$.** The proof of Theorem 4 says that QST:H "remains valid on $m(H)$": density operators and their von Neumann entropy are "left unchanged". As operators they change: moving them over through m gives $m\rho m^{-1}$, which is not even additive for the ordinary structure (`not_map_add_transport_projOp`). What should hold is that $\rho \mapsto m\rho m^{-1}$ is a bijection from the density operators of H onto those of $m(H)$, the space with the carried-over structure (`Twisted`), that keeps spectra and so entropy. This follows from `twistedIsometry` but is not formalized; nor is separability, which is trivial in dimension two. (Brief, Sec. 4, proof of Theorem 4.)

2. **Madelung representation.** "The Madelung equations ... provide such a representation", one "not even tied to vector spaces". The equivalence of the Madelung equations with quantum theory, and in what sense they represent QST:H, are not shown. (Brief, Sec. 5.)

From the formalization

1. None found. The blocker of the previous version of the brief (Theorem 1 as an independence statement) is resolved: the revised brief states its mathematical content as Proposition 1 and Corollary 3, which are checked in Lean, and argues Theorem 4 from them informally, which is checked by reading (Log, "Differences from the brief").